# Mean test, σ known -> Z critical-value method

A1S2: SLIDES 1-51 ONLY

## 1 RECOGNISE IT

- Hypothesis test about population mean μ.
- Population σ known.
- Asked for critical value/rejection decision; here use the in-scope two-tail route.

## 2 DECISION / METHOD

1. State $H_0$/$H_1$ and α.
2. Check Z lane: σ known.
3. SE=σ/√n; compute zcalc.
4. Two-tail criticals use α/2.
5. Reject if zcalc is outside the criticals; conclude in context.

## 3 FORMULA

$Zcalc=(\bar{X}-\mu_0)/(\sigma/\sqrt{n})$

$z^* = \Phi^{-1}(1-\alpha/2) > 0$

two-tail criticals = $\pm z^*$

reject if $|Zcalc| > z^*$

## 4 FORMULA SHEET VS WHAT I KNOW

Gives: Z lane and SE=σ/√n.
Still know: test statistic, critical values, decision rule and wording.

## 5 EXCEL ROUTE

- n=COUNT(range); Xbar=AVERAGE(range)
- criticals: NORM.S.INV(alpha/2), NORM.S.INV(1-alpha/2)
- zcalc=(X̄-μ₀)/(σ/SQRT(n)).

## 6 TINY WORKED EXAMPLE

$H_0$:μ=100, $H_1$:μ≠100, α=.05, X̄=104,σ=12,n=36. zcalc=2.00; |2|>1.96 -> reject $H_0$.

## 7 TOP EXAM TRAPS

- Using sample s despite supplied σ.
- Comparing X̄ directly with a Z critical.
- Using one critical for a two-tail test.
- Saying reject because z is positive.

## 8 SAY ALOUD

Known σ means standardise X̄ with σ/√n, then compare zcalc with the rejection boundary.

## 9 EXIT CHECK

If zcalc=1.4 in a two-tail α=.05 Z test, what is the decision?

**ONE-GLANCE VISUAL ROUTE**

